A COUNTEREXAMPLE TO GENERIC INJECTIVITY OF THE TANGENT GAUSS MAP

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We work over $\mathbb{C}$. For a curve $C \subset A$ in an abelian variety, the tangent Gauss map sends a smooth point to its tangent line, translated to the origin:

$$\phi_C : C^{\mathrm{sm}} \dashrightarrow \mathbb{P}(T_0 A), \qquad p \longmapsto d(t_{-p})_p(T_p C).$$

We call C nondegenerate if it is not contained in a translate of a proper abelian subvariety.

Theorem 1. *There exists a nondegenerate curve in an abelian threefold whose tangent Gauss map has generic degree 3.*

1. Strategy

Let C be a smooth non-hyperelliptic curve of genus 4, and let $\mathcal{L}$ be a line bundle of degree 3 with $h^0(\mathcal{L}) = 2$. It is base-point-free, since a base point would give a degree-two map to $\mathbb{P}^1$. Thus $|\mathcal{L}| : C \longrightarrow \mathbb{P}^1$ has degree 3. The linear system $(\mathcal{L}^{\otimes 2}, \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}))$ defines the map

$$\phi : C \xrightarrow[3:1]{|\mathcal{L}|} \mathbb{P}^1 \xhookrightarrow{\nu_2} \mathbb{P}^2,$$

where ν_2 is the quadratic Veronese embedding. In particular, ϕ has generic degree 3 onto its image and is not generically injective.

To obtain a counterexample by this construction, we seek a pair $(C, \mathcal{L})$ for which this map is a tangent Gauss map. It suffices to impose two further conditions:

(1) $\mathcal{L}^{\otimes 2} \cong \omega_C$;
(2) $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) \subset \mathrm{H}^0(\omega_C)$ is the $(1,0)$-part of a rational Hodge substructure of $\mathrm{H}^1(C; \mathbb{Q})$.

The second condition allows these three canonical forms to be pulled back from invariant forms on an abelian threefold; we make this passage in Section 2.4.

If we temporarily drop the nonhyperellipticity requirement, both conditions can be met. Take a smooth hyperelliptic curve C_0 of genus 4 admitting a tangential cover $\pi_0 : (C_0, p_0) \longrightarrow (E_0, 0)$ at a Weierstrass point p_0. Such examples are supplied by the Lamé spectral curves [3, Table 3 and Theorem L] and [2, Proposition 3.1]. Tangentiality means that $\pi_0^* E_0 \subset \mathrm{Jac}(C_0)$ is tangent to the Abel–Jacobi image at p_0. Put $\mathcal{L}_0 = \mathcal{O}_{C_0}(3p_0)$. Then

$$\mathcal{L}_0^{\otimes 2} \cong \omega_{C_0}, \qquad \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_0) = \mathrm{H}^0\left(\omega_{C_0}(-2p_0)\right).$$

Indeed, a hyperelliptic coordinate x with a double pole at p_0 gives the basis $1, x$ of $\mathrm{H}^0(\mathcal{L}_0)$, with products $1, x, x^2$. Tangentiality gives the Hodge condition, as explained in Lemma 2.1. This seed satisfies the two conditions, but $\mathcal{L}_0$ has a base point at p_0. We therefore deform the pair while preserving both conditions and leaving the hyperelliptic locus.

2. A dimension count for deformations

We first preserve the spin condition and the two sections, then impose compatibility of two different deformations of the cohomology subspaces. A dimension comparison will allow us to remove the hyperelliptic locus.

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2.1. **The deformation space and the spin condition.** Let B be a small simply connected marked deformation space of the spin curve $(C_0, \mathcal{L}_0)$. Thus $\mathcal{L}_t^{\otimes 2} \cong \omega_{C_t}$ throughout B, and $\dim B = 3g - 3 = 9$. To retain two sections, restrict to the theta-null locus

$$S = \{t \in B \mid h^0(\mathcal{L}_t) \geqslant 2\}$$

locally defined by one theta constant. Hence $\dim S \geqslant 8$. After shrinking B, upper semicontinuity gives $h^0(\mathcal{L}_t) = 2$ on S.

2.2. **Compatibility of the two deformations.** At the seed, set

$$V_0 := \left(\pi_0^* \mathrm{H}^1(E_0; \mathbb{Q})\right)^{\perp} \subset \mathrm{H}^1(C_0; \mathbb{Q}),$$
$$W_0 := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_0) \subset \mathrm{H}^0(\omega_{C_0}),$$

where $\perp$ is taken with respect to the cup-product pairing. Tangentiality gives $V_{0,\mathbb{C}} = W_0 \oplus \overline{W}_0$. Over S, we deform these subspaces in two different ways:

$$V_0 \rightsquigarrow V_t \subset \mathrm{H}^1(C_t; \mathbb{Q}) \quad \text{by Gauss–Manin parallel transport,}$$
$$W_0 \dashrightarrow W_t := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) \quad \text{by deforming the line bundle.}$$

Thus V_t is a rational subspace of dimension 6, whereas $W_t \subset \mathrm{H}^0(\omega_{C_t})$ is a holomorphic subbundle of rank 3. These constructions need not remain compatible: V_t may no longer be a Hodge substructure, and W_t need not lie in its complexification. Their compatibility is precisely the condition we impose:

$$\begin{aligned}
&V_t \text{ remains a Hodge substructure} \\
\iff\ & V_{t,\mathbb{C}} \cap \mathrm{H}^0(\omega_{C_t}) = W_t \\
\iff\ & V_{t,\mathbb{C}} = W_t \oplus \overline{W}_t \\
\iff\ & W_t \subseteq V_{t,\mathbb{C}} \\
\iff\ & \mathrm{rk}\left[W_t \longrightarrow \mathrm{H}^1(C_t; \mathbb{C})/V_{t,\mathbb{C}}\right] = 0.
\end{aligned}$$

The middle equivalence follows from the rationality of V_t and Hodge type, since $\dim W_t = 3$ and $\dim V_t = 6$. The last condition is the vanishing of a bundle map of ranks 3 and 2, so it imposes at most six equations on S. Consequently, every irreducible component Z of its zero locus through the seed satisfies

$$\dim Z \geqslant 9 - 1 - 6 = 2.$$

Fix one such component Z.

2.3. **Removing the hyperelliptic locus.** To find a nonhyperelliptic point, it remains to bound $\dim(Z \cap \mathcal{H}_4)$, where $\mathcal{H}_4 \subset B$ is the hyperelliptic locus. The following lemma reduces this to counting tangential covers.

Lemma 2.1. *Let C be a smooth hyperelliptic curve of genus 4, and let $\mathcal{L}^{\otimes 2} \cong \omega_C$ with $h^0(\mathcal{L}) = 2$.*

1. *There is a Weierstrass point p such that $\mathcal{L} \cong \mathcal{O}_C(3p)$.*
2. *For this point p, the subspace $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) = \mathrm{H}^0(\omega_C(-2p))$ is the $(1,0)$-part of a rational Hodge substructure if and only if C admits a tangential cover at p.*

Proof.

(1) Fix a Weierstrass point p_0. If $\mathcal{L}$ were base-point-free, its degree-three map and the degree-two map $|\mathcal{O}_C(2p_0)|$ would give a birational map to a curve of bidegree $(2,3)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, of arithmetic genus 2. Thus $\mathcal{L}$ has a base point p, and uniqueness of the degree-two linear system gives $\mathcal{L}(-p) \cong \mathcal{O}_C(2p_0)$. We get

$$\mathcal{O}_C(6p_0) \cong \omega_C \cong \mathcal{L}^{\otimes 2} \cong \mathcal{O}_C(4p_0 + 2p).$$

Hence p is Weierstrass and $\mathcal{L} \cong \mathcal{O}_C(3p)$.

(2) Under $T_0^* \operatorname{Jac}(C) = \mathrm{H}^0(\omega_C)$, the space $\mathrm{H}^0\left(\omega_C(-2p)\right) = \mathrm{H}^0\left(\omega_C(-p)\right)$ is the annihilator of the tangent line at p. Its being the $(1,0)$-part of a rational Hodge substructure is therefore equivalent to this line being the tangent space of an elliptic subvariety of $\operatorname{Jac}(C)$. Dualizing the inclusion and composing with the Abel–Jacobi map gives a tangential cover; the converse follows from the same annihilator description.

□

Consequently, the hyperelliptic points of Z give tangential covers $\pi_t : (C_t, p_t) \longrightarrow (E_t, 0)$. The complementary Hodge substructure $V_t^{\perp}$ has rank 2 and determines E_t up to isogeny. Choosing its saturated integral lattice fixes the pullback map on integral cohomology under local markings. The cup-product pairing then shows that $\deg \pi_t$ is locally constant.

Remark 2.1. *For a fixed elliptic curve and degree, there are only finitely many hyperelliptic tangential covers* [4, Proposition 2.5, 3.; Section 2, Remark 1, 3.–4.].

Since elliptic curves have one modulus and the choices of Weierstrass points p_t are finite,

$$\dim\left(Z \cap \mathcal{H}_4\right) \leqslant 1 < 2 \leqslant \dim Z.$$

We can therefore choose a nonhyperelliptic point of Z.

2.4. **Passing to a curve in an abelian threefold.**

Conclusion of the proof of Theorem 1.

(1) *Construction.* For the chosen point, the rational Hodge substructure V gives a quotient $q : \operatorname{Jac}(C) \longrightarrow A$ with connected kernel and $q^* \mathrm{H}^0(\Omega^1_A) = \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L})$. Composing this map with the Abel–Jacobi map gives $f : C \longrightarrow A$.

(2) *Immersion and nondegenerativity.* The pulled-back invariant forms are the products of sections of $\mathcal{L}$, so f is immersive and its tangent map is ϕ. Its image contains the origin and generates A, hence is nondegenerate.

(3) $\deg f = 1$. Since ϕ_C factors through f, we have $\deg f \mid 3$. If $\deg f = 3$, then the normalization of $\operatorname{Im} f$ is isomorphic to $\mathbb{P}^1$, but an abelian variety cannot contain a subvariety with geometric genus 0. Therefore C is the normalization of $f(C)$ and the tangent Gauss map of $f(C)$ has generic degree 3.

□

3. Simplicity and symmetries

Theorem 2.

(1) *The abelian threefold in Theorem 1 can be chosen simple.*

(2) *For this choice, the resulting curve is preserved by no nonidentity automorphism of A with scalar differential.*

Proof.

(1) We can choose a seed with simple quotient. Take $E_0 : y^2 = 4x^3 - 4x - 1$, let e_1, e_2, e_3 be the roots of $4e^3 - 4e - 1$, and take the smooth projective model of

$$C_0 : \quad \nu^2 = (B^3 - 208B + 560) \prod_{i=1}^{3} \left(B^2 + 10e_i B - 35e_i^2 - 28\right).$$

This is the genus-four Lamé spectral curve, with a degree-ten cover of E_0 tangential at infinity [3, Table 3 and Theorem L] and [2, Proposition 3.1 and the discussion after (17)]. The cover is defined over $\mathbb{Q}$, and E_0 has discriminant 37. Modulo 11, the polynomial on the right is the square-free polynomial $B^9 + B^4 + 6B^3 + 9B^2 + 7B + 5$; hence C_0 has genus 4 and good reduction. Direct counting gives

$$(\#C_0(\mathbb{F}_{11^r}))_{r=1}^{4} = (10, 120, 1312, 15312), \qquad \#E_0(\mathbb{F}_{11}) = 7.$$

Newton's identities and division by the elliptic factor $T^2 - 5T + 11$ give the Frobenius polynomial of $A_0 = \mathrm{Jac}(C_0)/\pi_0^* E_0$:

$$h(T) = T^6 + 3T^5 + 5T^4 - 14T^3 + 55T^2 + 363T + 1331.$$

It is irreducible modulo 3, and $11 \nmid 14$, so this reduction is simple and ordinary. An ordinary simple abelian threefold over a finite field of odd cardinality is absolutely simple unless its Frobenius polynomial has the form $T^6 + cT^3 + q^3$ [1, Proposition 3.4]. Thus this reduction is absolutely simple, and so is A_0 in characteristic zero, since a proper abelian subvariety would specialize to one after a finite extension.

The nonsimple locus is a countable union of proper irreducible analytic subsets of Z, so a very general point gives a simple quotient.

(2) Suppose $\tau \in \mathrm{Aut}(A)$ preserves $f(C)$ and has scalar differential. It lifts to an automorphism σ of C with $\phi_C \circ \sigma = \phi_C$, so $\mathrm{ord}(\sigma) \mid 3$. If $\sigma = \mathrm{id}$, then τ fixes a generating curve pointwise and therefore $\tau = \mathrm{id}$. If $\mathrm{ord}(\sigma) = 3$, then σ permutes the fibers of ϕ_C, so $C/\langle\sigma\rangle \cong \mathbb{P}^1$. Since τ^3 fixes the generating curve pointwise, $\tau^3 = \mathrm{id}$. Thus τ is either a translation or a rotation, by which we mean here an automorphism whose linear part is multiplication by a primitive cube root of unity.

- If τ is a translation, then ϕ_C is unramified, which contradicts the Riemann–Hurwitz formula;
- If τ is a rotation, then A is isogenous to $E_\rho^{\oplus 3}$ by [5, Lemma 3.15], which contradicts the assumption that A is simple.

□

References

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